

RESEARCH INTRODUCTION · WEEK 1

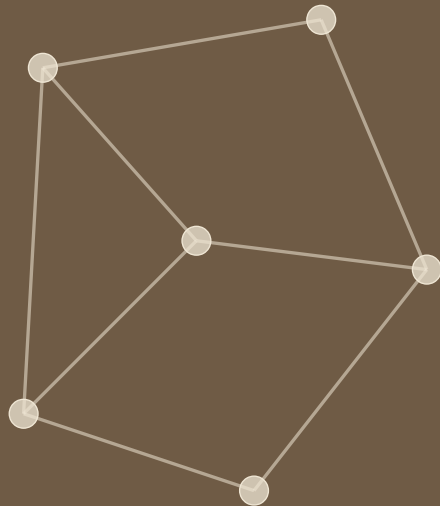
Formalizing Flag Algebras in Lean 4

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Lean 4

Graph Theory

Formalization



Flag Algebras, Formalized

THE PROBLEM

Extremal graph problems

How dense can a graph be without containing a forbidden pattern?

e.g. no triangles $\rightarrow$ edge density $\leq 1/2$



THE METHOD

Flag algebras (Razborov, 2007)

Reduce extremal graph problems to optimization problems that computers can solve

Flag Algebras, Formalized

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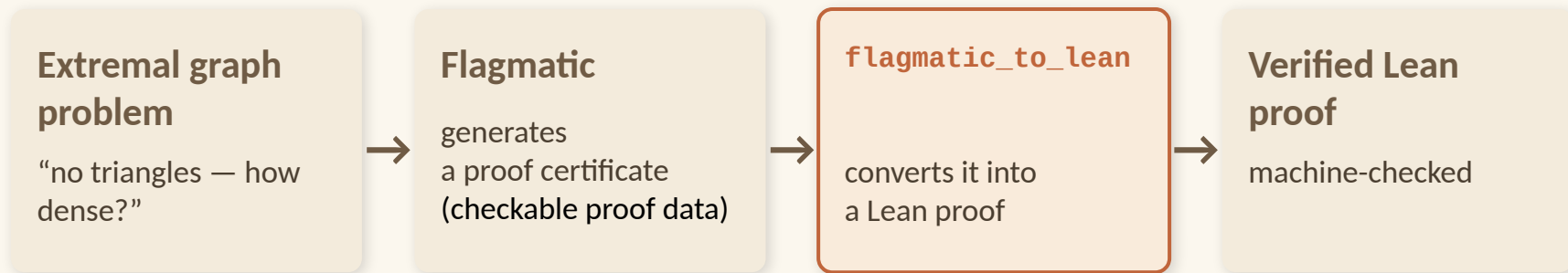
Flag algebras (Razborov, 2007)

Reduce extremal graph problems to optimization problems that computers can solve

OUR WORK

- **Formalized the flag algebra framework in Lean 4**
- A reusable Lean library for extremal graph theory
- Foundation for certificate-based verification

From Certificate to Verified Proof



OUR WORK

Automatically converts Flagmatic certificates into verified Lean proofs.

Automatically Generated Proof (Mantel's Theorem)



triangle-free $\implies$ edge density $\leq 1/2$

Generated automatically from a Flagmatic certificate:

```
theorem mantel_flagAlgebra :  
  FlagAlgebra_2_0_0_1 ≤[completeGraph (Fin 3)]  
    (1/2 : ℝ) • 1 := by  
  apply forbidLEWith_add_QuadraticForm ...  
  simp [flagQuadraticForm, v, M_real, ...]  
  ⋮  
  flag_nonneg
```

ALREADY USED FOR

- Mantel
- Turán K_4
- Turán K_5
- Erdős pentagon
- ...

Using AI in My Work

WHEN I STARTED LLMs weren't reliable enough for Lean development.

RECENTLY They are too useful to ignore.

BY HAND

I proved the first problems manually.
The same certificate-based proof pattern kept repeating.

Working with LLMs

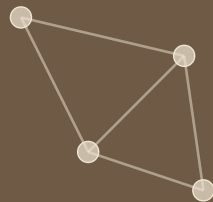
“This structure repeats — how can we automate it?”

We discussed the design. The LLM helped implement it
: metaprogramming, custom tactics, proof-generating scripts

NOT JUST TOOLING

LLMs can also help with mathematical reasoning — e.g. showing that our Lean formalization is compatible with Razborov's framework.

Going Forward



WORKING WITH AI

LLMs don't just save time — they sometimes open routes I wouldn't have tried on my own.

LLMs are becoming less a coding assistant and more a research collaborator.

LOOKING AHEAD — WHAT I'D LIKE TO EXPLORE

Improve the system

- refactor early proofs
- optimize the tactics used in the proofs

Expand the framework

- generalize flag algebras beyond simple graphs — to richer combinatorial objects

Step by step, toward machine-checked extremal graph theory.

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Thank you!